Hook and loop counts

How Velcro Works

Name _____ Date _____

Show your thinking with numbers, marks, or words.

1. The engineer counts 48 hooks on one burr and 84 hooks on another. Put >, <, or = in the circle. What is the greater count (or the shared count if equal)?

48 ○ 84

Show your thinking.

2. Little Wonderer counts 63 hooks in one tiny square and 36 loops in another. Put >, <, or = in the circle. What is the greater count (or the shared count if equal)?

63 ○ 36

Show your thinking.

3. The big one counts 55 loops on one patch and 55 loops on a second patch. Put >, <, or = in the circle. What is the greater count (or the shared count if equal)?

55 ○ 55

Show your thinking.

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4. Big Wonderer peels the shoe strap open 27 times. What is 27 rounded to the nearest ten?

27 → nearest ten: ___

Show your thinking.

5. The bag hangs on the strip for 74 seconds before the little one peels it. What is 74 rounded to the nearest ten?

74 → nearest ten: ___

Show your thinking.

6. A comb lifts 45 bits of lint out of an old strip. What is 45 rounded to the nearest ten?

45 → nearest ten: ___

Show your thinking.

WONDER DAILY

How Velcro Works

A strap that rips

A shoe strap rips open, then presses shut and holds. Its plain name is hook and loop: one rough side, one soft.

A burr on a walk

In 1941, burrs stuck to an engineer's socks and his dog. Under a microscope, each spine ended in a tiny hook.

Hooks meet loops

Stiff hooks on one side catch soft loops on the other. One hook is weak, but hundreds together hold strong.

Pull or peel?

A sideways pull is hard, because every hook pulls at once. A peel is easy: only one thin line lets go.

A bag that hangs

A hanging bag pulls along the strip, so every hook shares the weight. Peel one corner, and the bag slides off.

When grip fades

Nylon loops were heated, then snipped into hooks. Lint clogs old hooks, but a fine comb can lift it out.

Make a little book

1 Fold it into 8 boxes. Open it flat.

2 Fold short end to short end. Cut the thick line. Open it.

3 Fold the long way, pictures out. Push the ends in.

4 Fold into a book. Page 1 goes on the front!

Explain a hook clue in your words

How Velcro Works · Curiosity day

Name _____ Date _____

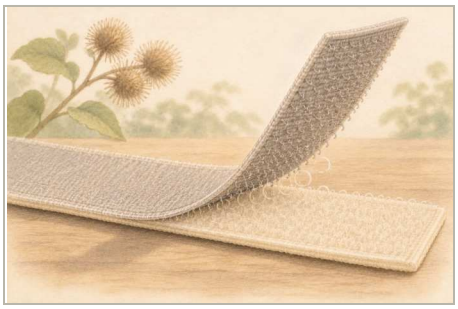
Choose one picture. Say your idea first, then draw or write it.



A burr on a walk



Hooks meet loops



Pull or peel?

Draw one hook and one loop. Show the hook catching the loop.

hook · loop · burr · peel · because · maybe

The idea came from a burr because … what was on its spines?

The strips hold together because … what do the hooks catch?

A peel opens easily because … how many hooks let go at once?

Does your sentence tell what you noticed and use because or maybe to explain?



A Thousand Tiny Hands

Little Wonderer opened a strap on a shoe,
and it went RRRIP! like a grumpy kazoo.
"Who's hiding in there?" asked the small one, aghast.
"A thousand tiny hands," said the big one, "holding fast."

One side is all hooks, like a crowd of bent canes,
or umbrellas turned upside down, left out in the rains.
The other side's loops, like a soft curly mat,
and the hooks grab the loops, and that's simply that.

Big Wonderer said, "Slide them sideways — they'll hold,
every hook pulling hard, all together and bold.
But peel from a corner, a few at a time,
and they let go as easy as saying this rhyme."

Who dreamed it up first? Not a wizard or king,
but a burr on a dog, such a scratchy small thing.
So the little one bowed to a burr on its stem:
"Thank you for holding on. We learned it from them."